MILITARY SPECIFICATION

RACK, TEST TUBE, LABORATORY, Wasserman,

Corrosion-Resisting Steel, 20 Tube

This limited coordination Military Specification has been prepared by the Defense Personnel Support Center based upon currently available technical information. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers a twenty-tube Wasserman metal test tube rack suitable for use with water baths (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids or request for porposals form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Setup.

Military

MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof, Flexible.

STANDARDS

Federal

FED-STD-66	Steel, Chemical Composition and Hardenability.

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Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The test tube rack shall be fabricated from .025 inch thick No. 24 United States Standard (USS) gauge corrosion-resisting steel that conform to the requirements of class 302 or 304 of FED-STD-66. Commercial tolerances on the thickness shall apply.

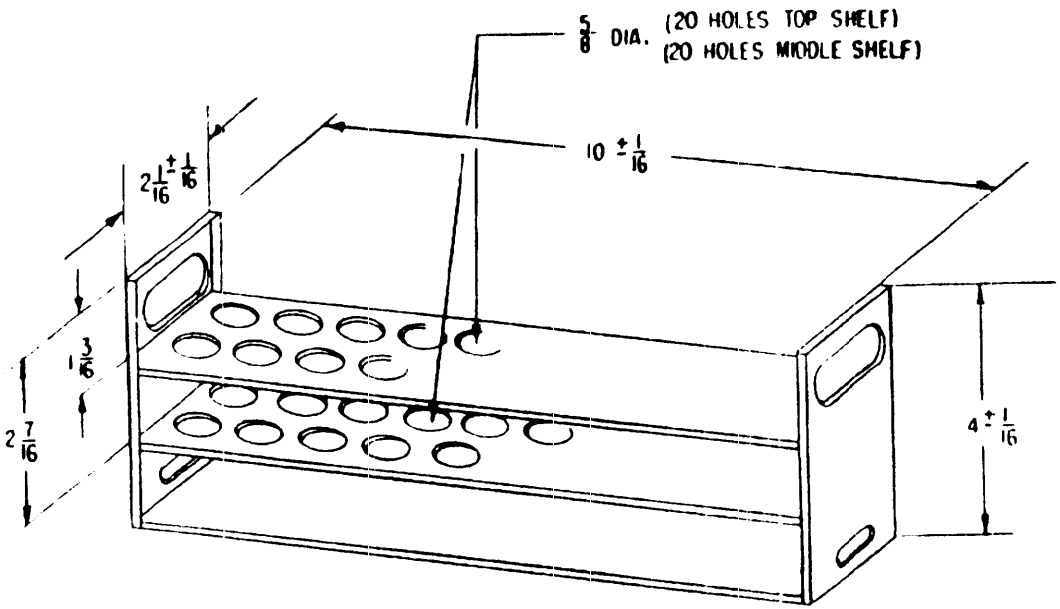
3.2 Design and construction. The test tube rack shall conform to Figure 1 for design and dimensions. The test tube rack shall consist of: two perforated flanged shelves, an unperforated flanged base, and supporting ends with cutouts for lifting and drainage. The shelves shall have two rows of ten holes each; and the center lines of corresponding holes in the two shelves shall be in proper register to assure that inserted tubes rest vertically in the rack. The holes shall be numbered in pairs from 1 to 10 lengthwise, with legible impressions along the center of the top shelf between the two rows of holes. The component base, shelves and ends of the test tube rack shall be formed on one-piece sheet metal. The ends of the shelves and base shall be flanged a minimum of 1/4 inch. The flanged ends shall be fitted inside the flanged end walls and affixed by not less than three spot welds at each end.

3.3 Finish. Weld marks on end supports shall be polished to blend with adjacent surfaces. Weld indentations are permissible. All surfaces shall be uniformly finished to a No. 4 commercial finish.

3.4 Identification marking. Each rack shall be permanently and legibly marked with the following: Federal Stock Number, contract or purchase order number, name or registered trademark of the manufacturer, and letters "U.S.". Marking secured by adhesives shall not be permitted.

3.5 Workmanship. Workmanship shall be first class throughout. The welding shall be free of cracks or porosity. The racks (and cutouts) shall be free from sharp edges, burrs, cracks, dents, pits, and other defects which detract from their appearance or may impair their serviceability. The racks shall be free of rocking or wobbling when resting on a flat surface.

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NOTE:

1 - ALL DIMENSIONS ARE IN INCHES.

2 - THE TOLERANCE ON THE 5/8 INCH DIAMETER HOLES IS -0 + 1/64.

3 - UNLESS OTHERWISE INDICATED THE TOLERANCE ON ALL DIMENSIONS IS ± 1/32.

4 - FREE EDGES OF SHELVES, BASE, AND END WALLS FLANGED 5/16 ± 1/16 INCH AND HEMMED 1/8 INCH MINIMUM.

TEST TUBE RACK

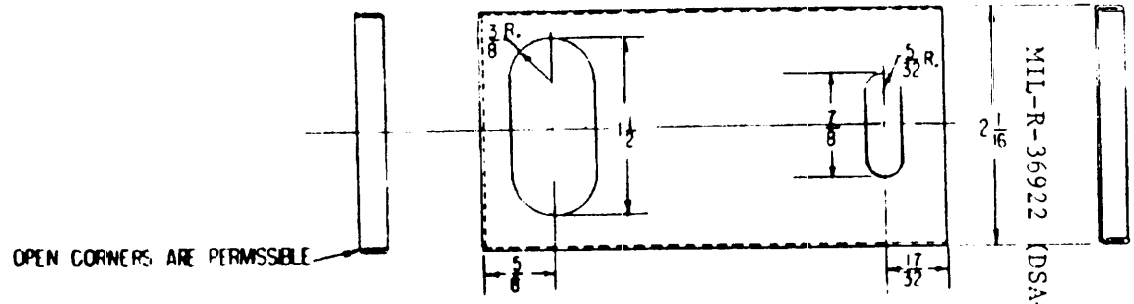


FIGURE 1

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or nay other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractir and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, or auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table 1. The unit of product for examination purposes shall be the individual test tube rack.

Table I. For examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-2	2.5

4.3 Tests. Tests shall be conducted as necessary to determine compliance with specification requirements.

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4.4 Examination. The test tube racks shall be examined to determine compliance with all requirements contained in this specification.

4.4.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects indicated in Table II. Examination shall not be restricted to the classified possible defects listed in the table.

Table II. Classification of defects

Categories	Defects
Major	
101	Rack (including cutouts) not free of sharp edges.
102	Rack not free of dents or distortion
103	Shelves not properly secured to end walls.
104	Welds not sound or not free of cracks or porosity.
105	Rack not free of corrosion.
106	Rack not free of rocking or wobbling when resting on a flat surface.
107	Test tube holes on shelves not properly aligned.
Minor	
201	Rack not free of pits or burrs.
202	Rack not free of dirt, grease, metallic particles or other foreign matter.
203	Welds not smooth.
204	Finish not as specified or not uniform on exposed surfaces.
205	Exposed surfaces not free of scratches, tooling or grinding marks.
206	Identification marking not complete or legible.
207	Test tube numbers on rack not complete or legible.

4.4.2 Dimensional examination. The test tube racks shall be examined for defects in dimensions. Any dimension not within specified limits shall be classified as a defect.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Unit of issue. One rack, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table III shall be packaged in each unit intermediate and exterior container, as applicable, for the required level of protection specified in the procurement document.

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Table III. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	36 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 Box.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a water-proof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Special requirements of applicable table shall apply for PPP-B-636 Box.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The test tube rack is intended to be suitable for use with serological water baths conforming to requirements of latest revision of GG-W-115 and the following stock listed water baths: FSN's 6640-765-0599, 6640-765-0621 and 6640-765-0680.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Federal Stock Number (FSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification or those which are commercially available, but is intended to cover the type normally procured to meet Military requirements.

6.4 This specification covers the following item shown in the Federal Supply Catalog:

Federal Stock Number	Item Identification
6640-442-6450	RACK, TEST TUBE, LABORATORY, Wasserman, Corrosion-Resisting Steel, 20 Tube

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Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities
Army - MD
Navy - MS
Air Force - 03

Preparing activity
Defense Supply Agency - DM

Project No. 6640-0772

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-R-36922 (DSA-DM)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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